

AMPL Training Series · Part 2 of 2



AMPL

The Modern AMPL Stack

*From the legacy IDE to Python-first workflows
for teaching, research, and the road to industry:
Hands-on session*

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THE HOUR AHEAD

Today's Agenda

01 **Part One Summary**

02 **Transitioning Workflows and Live Demo**

Quick checklist

- ☐ Python installed.
- ☐ VS Code installed.
- ☐ Academic license activated (UUID).
- ☐ Sample model and run files ready.

SECTION 01

Part One Summary

Toolkit

VS Code + AMPL extension

Syntax highlighting, debugging, integrated terminal, Git — the editor your students already know.

AMPL + solvers distributed as python packages via pip.

Jupyter Notebook + Colab

Narrative + code + output in one artifact. Zero-install via Colab kills the 'doesn't work on my machine' problem.

MO-Book + amplbot

Free, Colab-runnable optimization textbook + AI assistant trained on AMPL syntax.

Use as much or as little of the modern stack as fits your course.

AMPL Licenses for Academia

<https://ampl.com/academia/>

A4A Academics

- Anyone with academic email
- 10 commercial + open source solvers
- No time limit, 12 months renewal
- Self-serve in Portal
- Free, individual use

A4C Courses

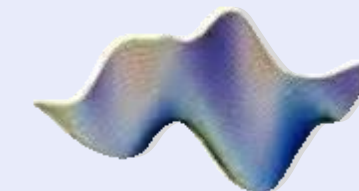
- Instructor-requested, distributable
- All commercial solvers, no fees
- Up to 8 months
- Shareable activation link
- Covers the whole class

<https://portal.ampl.com/user/ampl/request/ampl-for-courses/new>

ACS

- Recent grads (≤ 24 months)
- All commercial solvers
- 12 months · commercial OK
- Bridges classroom → industry

<https://portal.ampl.com/user/ampl/request/ampl-career-starter/new>



Activate A4A

01 · Open this URL

portal.ampl.com/account/ampl/

02 · Sign up with your academic email

Verify the link in your inbox — may take 30–60 seconds.

03 · Generate / locate your UUID

Copy it. You'll paste it in the following sections.

SECTION 02

Transitioning Workflows And Live Demo

Install amplpy + activate.

COLAB

- Go to <https://github.com/ampl/modern-ampl> and upload notebook to colab.
- Alternatively, go directly https://colab.research.google.com/drive/1nPExSwNZnNwbkkcKPMFp0TyMnhaO_2fs?usp=sharing

VS CODE + Extension

- ✓ Install python from <https://www.python.org/>
- ✓ Install VS Code from <https://code.visualstudio.com/>
- Install the **AMPL (Official)** extension from the Marketplace.
- Other recommended extensions
 - **Python**
 - **Jupyter**
 - **Colab**
- Run commands on the right to activate license.

COMMON GOTCHAS

- python vs python3 — stick with one
- Corp network/laptop blocking pip → fall back to Colab.

```
~/TERMINAL

$ python3 -m venv .venv
$ source .venv/bin/activate
(.venv) $ python -m pip install amplpy pandas matplotlib ipykernel openpyxl
(.venv) $ python -m amplpy.modules install highs gurobi cplex xpress
(.venv) $ python -m amplpy.modules activate <paste-your-uuid>
(.venv) $ python -m amplpy.modules run ampl -vvq
```

After running the commands, go to the notebook and follow the demo.

RESOURCES

Where to go from here.

01

AMPL forum

discuss.ampl.com

Where the “I got stuck after the session” questions live. Active community, fast turnaround.

02

MO-Book

ampl.com/mo-book

Free, Colab-runable optimization textbook. A ready-made library of teachable notebooks.

03

Solver docs

dev.ampl.com/solvers

Per-solver options, MP-driver flags, quirks worth knowing before assigning a problem.

04

Academic questions

academia@ampl.com

Licensing, A4A / A4C / ACS.

AI ASSISTANTS FOR AMPL

AMPL LLM Offerings



Google Gemini Gem
gemini.google.com



ChatGPT custom GPT
chatgpt.com/g/amplbot



Claude Skill
github.com/ampl/claude-for-ampl



NotebookLM
notebooklm.google.com

Questions?

<https://github.com/ampl/modern-ampl>

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